

Anti-inflammatory activity of natural dietary flavonoids.



Over the past few decades, inflammation has been recognized as a major risk factor for various human diseases. Acute inflammation is short-term, self-limiting and it's easy for host defenses to return the body to homeostasis. Chronic inflammatory responses are predispose to a pathological progression of chronic illnesses characterized by infiltration of inflammatory cells, excessive production of cytokines, dysregulation of cellular signaling and loss of barrier function. Targeting reduction of chronic inflammation is a beneficial strategy to combat several human diseases. Flavonoids are widely present in the average diet in such foods as fruits and vegetables, and have been demonstrated to exhibit a broad spectrum of biological activities for human health including an anti-inflammatory property. Numerous studies have proposed that flavonoids act through a variety mechanisms to prevent and attenuate inflammatory responses and serve as possible cardioprotective, neuroprotective and chemopreventive agents. In this review, we summarize current knowledge and underlying mechanisms on anti-inflammatory activities of flavonoids and their implicated effects in the development of various chronic inflammatory diseases. This journal is © The Royal Society of Chemistry 2010